Geospatial Data Analysis

Regional Market Expansion Opportunities in India

1. Executive Summary

The supplied dataset contains 400 original records. After geographic validation and identifier checks, 400 records remain for analysis. The analysis compares search demand, competition, conversion rates, acquisition costs and relative market opportunity scores.

2. Methodology

The workflow includes dataset inspection, data cleaning, coordinate validation, state-level aggregation and interactive mapping using Folium. The screening score weights normalized demand at 50%, low competition at 30%, and conversion rate at 20%. Acquisition cost is assessed separately. This score is a relative screening metric and has not been validated as a profit prediction.

3. State-Level Market Summary

State	Regions	Total Demand	Avg Demand	Avg Competitors	Avg Conversion	Avg Acquisition Cost (INR)
Maharashtra	69	75,849	1,099.3	7.62	7.77%	470
Tamil Nadu	47	53,771	1,144.1	7.94	7.88%	412
Delhi	56	53,007	946.6	8.02	7.27%	486
Gujarat	50	52,110	1,042.2	8.38	8.40%	487
Telangana	42	48,103	1,145.3	7.19	8.15%	439
Kerala	39	39,015	1,000.4	6.79	8.07%	520
Rajasthan	33	34,849	1,056.0	7.67	7.98%	447
West Bengal	32	34,126	1,066.4	8.56	7.26%	392
Karnataka	32	33,789	1,055.9	7.72	8.78%	476

4. Top Expansion Candidates

City	State	Demand	Competitors	Conversion	Acquisition Cost (INR)	Score
Kochi	Kerala	1,816	1	13.81%	613	0.973

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Kolkata	West Bengal	1,799	1	13.55%	727	0.964
Kochi	Kerala	1,787	1	13.05%	407	0.952

5. Geographic Visualization

The accompanying interactive HTML map displays supplied location coordinates and market indicators. Open regional_market_map.html in a browser to inspect individual records and their opportunity scores.

6. Limitations and Recommendations

The supplied data is synthetic and contains no direct sales or revenue measurements. Search demand is used as a demand proxy, while competitor counts are used as a competition proxy. The coordinates have been checked against broad geographic bounds, but not independently verified against city boundaries. Before expansion, validate local demand, actual customer acquisition costs, competitor coverage and expected profitability. Treat the ranked regions as candidates for further investigation rather than confirmed underserved markets.